

99.500 Applied Mathematics for Engineering Spring Semester AY2026

Ordinary Differential Equation

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Among all of the mathematical disciplines the theory of differential equations is the most important... It furnishes the explanation of all those elementary manifestations of nature which involve time.

AZ QUOTES

Week 3 Outline

- 1st-order linear ODE
- 2nd-order linear ODE
- Homogeneous vs non-homogeneous ODE
- Particular solution
- Mid-term project instruction

Definitions

ODE: a relation containing one real variable x , the real dependent variable y , and some of its derivatives y' , y'' , $\dots$, $y^{(n)}$, $\dots$, with respect to x . The order of an ODE: the order of the highest derivative that occurs in the equation. Thus, an n -th order ODE has the general form

$$F(x, y, y', y'', \dots, y^{(n)}) = 0$$

Linear ODE: the dependent variable y and all its derivatives appear to the power of one (are not multiplied or divided by each other) and are not composed with any non-linear functions of y or its derivatives. The general form of an n n-th order linear ODE is:

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_1(x)y' + a_0(x)y = g(x)$$

If $g(x) \equiv 0$, it is a **homogeneous** equation.

If $g(x) \not\equiv 0$, it is a **non-homogeneous** equation.

Condition for ODE solution

The exact solution of n -th order ODE requires n “constraints”:

1. Condition 1: If the independent variable is time, then the “constraints” are the initial conditions (ICs).
2. Condition 2: the independent variable is the position or location related, then the “constraints” are the boundary conditions (BCs).

1st-order linear ODE

General form of a 1st-order linear ODE: $a_1(x)y' + a_0(x)y = g(x)$

$$\begin{aligned} a_1(x)y' + a_0(x)y = 0 &\implies y' = -\frac{a_0(x)}{a_1(x)}y \\ &\implies \frac{dy}{y} = -\frac{a_0(x)}{a_1(x)}dx \\ &\implies \ln y = \int -\frac{a_0(x)}{a_1(x)}dx \\ &\implies y = Ce^{\int -\frac{a_0(x)}{a_1(x)}dx} \end{aligned}$$

Non-homogeneous ODE: Integrating Factor

$$a_1(x)y' + a_0(x)y = g(x) \implies y' + P(x)y = Q(x)$$

$$\text{Integrating factor :} \quad \mu(x) = e^{\int P(x)dx}$$

$$\frac{d}{dx}[\mu(x)y] = \mu(x)y' + \mu(x)P(x)y \quad \mu(x)y' + \mu(x)P(x)y = \mu(x)Q(x)$$

$$\implies \frac{d}{dx}[\mu(x)y] = \mu(x)Q(x)$$

$$\int \frac{d}{dx}[\mu(x)y]dx = \int \mu(x)Q(x)dx \implies \mu(x)y = \int \mu(x)Q(x)dx + C$$

$$\implies y = \frac{1}{\mu(x)} \left(\int \mu(x)Q(x)dx + C \right)$$

The solution to the 1st-order linear ODE $y' + P(x)y = Q(x)$ is:

$$y = \frac{1}{e^{\int P(x)dx}} \left(\int e^{\int P(x)dx} Q(x)dx + C \right)$$

Example

$$y' + 2y = e^x$$

Identify $P(x)$ & $Q(x)$: $P(x) = 2$ & $Q(x) = e^x$

Find the integrating factor $\mu(x)$: $\mu(x) = e^{\int 2dx} = e^{2x}$

Multiply the ODE by $\mu(x)$: $e^{2x}y' + 2e^{2x}y = e^{2x}e^x$

Recognize the left-hand side as a derivative : $\frac{d}{dx}(e^{2x}y) = e^{3x}$

Integrate both sides : $e^{2x}y = \int e^{3x}dx + C = \frac{1}{3}e^{3x} + C$

Solve for y : $y = \frac{1}{3}e^x + Ce^{-2x}$

2nd-order ODE: Method of Reduction to 1st Order

General form of a 2nd-order ODE : $y'' = f(x, y, y')$

y does not appear explicitly : $y'' + P(x)y' = Q(x)$

$$v = y' \text{ \& } v' = y'' \implies v' + P(x)v = Q(x)$$

$$\begin{aligned} \text{Method of integrating factor} &\implies v = \frac{1}{e^{\int P(x)dx}} \left(\int e^{\int P(x)dx} Q(x)dx + C \right) \\ &\implies y' = \frac{1}{e^{\int P(x)dx}} \left(\int e^{\int P(x)dx} Q(x)dx + C \right) \\ &\implies y = \int \frac{1}{e^{\int P(x)dx}} \left(\int e^{\int P(x)dx} Q(x)dx + C \right) dx \end{aligned}$$

Example

$$y'' - 3y' = x$$

$$v = y' \text{ \& } v' = y''$$

$$\implies$$

$$v' - 3v = x$$

Method of integrating factor

$$\implies$$

$$v = -\frac{1}{3}x - \frac{1}{9} + Ce^{3x}$$

$$\implies$$

$$y = \int -\frac{1}{3}x - \frac{1}{9} + Ce^{3x} dx$$

$$y = -\frac{1}{6}x^2 - \frac{1}{9}x + Ce^{3x} + D$$

2nd-order ODE: Method of Reduction to 1st Order

x does not appear explicitly: $y'' = f(y, y')$

$$\begin{aligned} v = y' &\implies y'' = \frac{dv}{dx} = \frac{dv}{dy} \frac{dy}{dx} = v \frac{dv}{dy} \\ &\implies v \frac{dv}{dy} = f(y, v) \text{ 1st-order ODE} \end{aligned}$$

$$\text{Find } v \implies y = \int v dx$$

Example

$$y'' + y = 0$$

$$\begin{aligned} v = y' &\implies v \frac{dv}{dy} = -y \\ &\implies \int v dv = - \int y dy \\ &\implies v^2 = -y^2 + C \\ &\implies v = \pm \sqrt{C - y^2} \\ &\implies \frac{dy}{dx} = \pm \sqrt{C - y^2} \\ &\implies y = \sqrt{C} \sin(A - x) + \sqrt{C} \sin(B + x) \\ &\implies y = c_1 \sin(x) + c_2 \cos(x) \end{aligned}$$

2nd-order Linear Homogeneous ODE with Constant Coefficients

$$y'' + ay' + by = 0$$

Look for solution of the form $y = e^{\lambda x}$

$e^{\lambda x}$ is a solution $\iff \lambda^2 + a\lambda + b = 0$

$$\lambda_1 = \frac{1}{2}(-a + \sqrt{a^2 - 4b})$$

$$\lambda_2 = \frac{1}{2}(-a - \sqrt{a^2 - 4b})$$

$$a^2 - 4b > 0 \implies y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$$a^2 - 4b = 0 \implies \lambda_1 = \lambda_2, y = c_1 e^{\lambda x} + c_2 x e^{\lambda x}$$

$$a^2 - 4b < 0 \implies \lambda_1 = \alpha + i\beta, \lambda_2 = \alpha - i\beta$$

$$\implies y = c_1 e^{(\alpha + i\beta)x} + c_2 e^{(\alpha - i\beta)x}$$

$$\implies y = e^{\alpha x} (c_1 e^{i\beta x} + c_2 e^{-i\beta x})$$

$$\implies y = e^{\alpha x} (c_1 \cos(\beta x) + ic_1 \sin(\beta x) + c_2 \cos(\beta x) - ic_2 \sin(\beta x))$$

$$\implies y = C e^{\alpha x} \cos(\beta x) + D e^{\alpha x} \sin(\beta x)$$

Examples

$$y'' + y' - 2y = 0, y(0) = 4, y'(0) = -5 \quad \text{Solution: } \lambda_1 = 1, \lambda_2 = -2$$
$$y = e^x + 3e^{-2x}$$

$$y'' - 4y' + 4y = 0, y(0) = 3, y'(0) = 1 \quad \text{Solution: } \lambda_1 = \lambda_2 = 2$$
$$y = (3 - 5x)e^{2x}$$

$$y'' - 2y' + 10y = 0 \quad \text{Solution: } \lambda_1 = 1 + 3i, \lambda_2 = 1 - 3i$$
$$y = e^x(c_1 \cos 3x + c_2 \sin 3x)$$

Example of Reaction in Batch Reactor

$$\begin{cases} \frac{d\tilde{C}_R}{d\tilde{t}} = -\tilde{C}_R + K_1\tilde{C}_I \\ \frac{d\tilde{C}_I}{d\tilde{t}} = \tilde{C}_R - K_1\tilde{C}_I - K_2\tilde{C}_I \\ \frac{d\tilde{C}_P}{d\tilde{t}} = K_2\tilde{C}_I \end{cases} \text{ with BCs } \begin{cases} \tilde{C}_R(0) = 1 \\ \tilde{C}_I(0) = 0 \\ \tilde{C}_P(0) = 0 \end{cases}$$

$$\frac{d\tilde{C}_I}{d\tilde{t}} = \tilde{C}_R - K_1\tilde{C}_I - K_2\tilde{C}_I \implies \tilde{C}_R = \frac{d\tilde{C}_I}{d\tilde{t}} + K_1\tilde{C}_I + K_2\tilde{C}_I$$

$$\text{Substitute } \tilde{C}_R \text{ into 1st ODE} : \quad \frac{d^2\tilde{C}_I}{d\tilde{t}^2} + (1 + K_1 + K_2)\frac{d\tilde{C}_I}{d\tilde{t}} + K_2\tilde{C}_I = 0$$

$$\lambda^2 + (1 + K_1 + K_2)\lambda + K_2 = 0$$

$$\lambda_{1,2} = \frac{-(1+K_1+K_2) \pm \sqrt{(1+K_1+K_2)^2 - 4K_2}}{2}$$

$$\tilde{C}_I = c_1 e^{\lambda_1 \tilde{t}} + c_2 e^{\lambda_2 \tilde{t}}$$

$$\tilde{C}_I(0) = c_1 + c_2 = 0$$

$$\tilde{C}_R(0) = \left. \frac{d\tilde{C}_I}{d\tilde{t}} \right|_{\tilde{t}=0} = c_1 \lambda_1 + c_2 \lambda_2 = 1$$

$$c_1 = \frac{1}{\lambda_1 - \lambda_2} \quad \& \quad c_2 = \frac{1}{\lambda_2 - \lambda_1}$$

$$\tilde{C}_I = \frac{e^{\lambda_1 \tilde{t}} - e^{\lambda_2 \tilde{t}}}{\lambda_1 - \lambda_2}$$

$$\tilde{C}_R = \frac{\lambda_1 e^{\lambda_1 \tilde{t}} - \lambda_2 e^{\lambda_2 \tilde{t}}}{\lambda_1 - \lambda_2} + (K_1 + K_2) \frac{e^{\lambda_1 \tilde{t}} - e^{\lambda_2 \tilde{t}}}{\lambda_1 - \lambda_2}$$

$$\frac{d\tilde{C}_P}{d\tilde{t}} = K_2 \tilde{C}_I = \frac{K_2 \left(e^{\lambda_1 \tilde{t}} - e^{\lambda_2 \tilde{t}} \right)}{\lambda_1 - \lambda_2}$$

$$\begin{aligned} \tilde{C}_P &= \frac{K_2}{\lambda_1 - \lambda_2} \int \left(e^{\lambda_1 \tilde{t}} - e^{\lambda_2 \tilde{t}} \right) d\tilde{t} \\ &= \frac{K_2}{\lambda_1 - \lambda_2} \left(\frac{e^{\lambda_1 \tilde{t}}}{\lambda_1} - \frac{e^{\lambda_2 \tilde{t}}}{\lambda_2} \right) + c_3 \end{aligned}$$

$$\tilde{C}_P(0) = \frac{K_2}{\lambda_1 - \lambda_2} \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right) + c_3 = 0$$

$$\Rightarrow c_3 = \frac{K_2}{\lambda_1 \lambda_2}$$

$$\tilde{C}_P = \frac{K_2}{\lambda_1 - \lambda_2} \left(\frac{e^{\lambda_1 \tilde{t}}}{\lambda_1} - \frac{e^{\lambda_2 \tilde{t}}}{\lambda_2} \right) + \frac{K_2}{\lambda_1 \lambda_2}$$

Nth-order Linear Homogeneous ODE with Constant Coefficients

General form : $y^{(n)} + a_1 y^{(n-1)} + \dots + a_{n-1} y' + a_n y = 0$

$y = e^{\lambda x}$ is a solution $\iff \lambda^n + a_1 \lambda^{n-1} + \dots + a_{n-1} \lambda + a_n = 0$

λ : characteristic value or eigenvalues

$$\lambda^n + a_1 \lambda^{n-1} + \dots + a_{n-1} \lambda + a_n = (\lambda - \lambda_1)^{m_1} (\lambda - \lambda_2)^{m_2} \dots (\lambda - \lambda_s)^{m_s}$$

$\lambda_1, \dots, \lambda_s$: distinct eigenvalues

$m_1, \dots, m_s$: multiplicity of respective eigenvalues

Theorem: Let $\lambda_1, \dots, \lambda_s$ be the distinct eigenvalues for **nth-order linear homogeneous ODE**, with multiplicity $m_1, \dots, m_s$ respectively. Then it has a fundamental set of solutions

$$\begin{aligned} &e^{\lambda_1 x}, x e^{\lambda_1 x}, \dots, x^{m_1-1} e^{\lambda_1 x}; \\ &\dots\dots\dots; \\ &e^{\lambda_s x}, x e^{\lambda_s x}, \dots, x^{m_s-1} e^{\lambda_s x}. \end{aligned}$$

The general solution for the homogeneous ODE is the linear combination of fundamental set of solutions: $y_h(x) = \sum_{i=1}^s \sum_{j=1}^{m_i} c_{ij} x^{j-1} e^{\lambda_i x}$

Nth-order Linear Non-homogeneous ODE

$$y^{(n)} + a_1(x)y^{(n-1)} + \cdots + a_{n-1}(x)y' + a_n(x)y = f(x)$$

The general solution is in the form: $y(x) = y_h(x) + y_p(x)$

homogeneous solution $y_h(x) : y_h^{(n)} + a_1(x)y_h^{(n-1)} + \cdots + a_{n-1}(x)y_h' + a_n(x)y_h = 0$

particular solution $y_p(x) : y_p^{(n)} + a_1(x)y_p^{(n-1)} + \cdots + a_{n-1}(x)y_p' + a_n(x)y_p = f(x)$

How to find the particular solution?

1. Method of variation of parameters
2. Method of undetermined coefficients

Method of Variation of Parameters

$$y'' + ay' + by = f(x)$$

Let y_1 & y_2 be two linearly independent homogeneous solutions

Look for $y_p = u_1 y_1 + u_2 y_2$

Suppose $u'_1 y_1 + u'_2 y_2 = 0$

Put y_p into $y'' + ay' + by = f \implies u'_1 y'_1 + u'_2 y'_2 = f$

$$\begin{cases} u'_1 y_1 + u'_2 y_2 = 0 \\ u'_1 y'_1 + u'_2 y'_2 = f \end{cases} \implies u'_1 = -\frac{y_2 f}{y_1 y'_2 - y'_1 y_2}, u'_2 = \frac{y_1 f}{y_1 y'_2 - y'_1 y_2}$$

$$u_1 = -\int \frac{y_2 f}{y_1 y'_2 - y'_1 y_2} dx$$

$$u_2 = \int \frac{y_1 f}{y_1 y'_2 - y'_1 y_2} dx$$

Example

$$y'' + y = \sec x$$

$$y_1 = \cos x, y_2 = \sin x$$

$$y_1 y_2' - y_1' y_2 = \cos x \cos x - (-\sin x) \sin x = 1$$

$$u_1 = -\int \sin x \sec x \, dx = \ln |\cos x| + c_1$$

$$u_2 = \int \cos x \sec x \, dx = x + c_2$$

$$y_p = \cos x \ln |\cos x| + x \sin x$$

$$y = c_1 \cos x + c_2 \sin x + \cos x \ln |\cos x| + x \sin x$$

Method of Undetermined Coefficients: Case 1

$$y'' + by' + cy = f(x)$$

$$f(x) = P_n(x)e^{\alpha x}$$

$P_n(x)$: polynomial of degree $n \geq 0$

Look for $y_p = Q(x)e^{\alpha x}$

$$Q'' + (2\alpha + b)Q' + (\alpha^2 + b\alpha + c)Q = P_n(x)$$

$$\left\{ \begin{array}{ll} \alpha^2 + b\alpha + c \neq 0 & \text{Choose } Q(x) = R_n(x), y = R_n(x)e^{\alpha x} \\ \alpha^2 + b\alpha + c = 0, 2\alpha + b \neq 0 & \text{Choose } Q(x) = xR_n(x), y = xR_n(x)e^{\alpha x} \\ \alpha^2 + b\alpha + c = 0, 2\alpha + b = 0 & \text{Choose } Q(x) = x^2R_n(x), y = x^2R_n(x)e^{\alpha x} \end{array} \right.$$

Examples

$$y'' - y' - 2y = 4x^2$$

$$\text{Solution: } y = c_1 e^{2x} + c_2 e^{-x} - 3 + 2x - 2x^2$$

$$y''' + 2y'' - y' = 3x^2 - 2x + 1$$

$$\text{Solution: } y = c_1 + c_2 e^{(-1+\sqrt{2})x} + c_3 e^{(-1-\sqrt{2})x} - 27x - 5x^2 - x^3$$

$$y'' - 2y' + y = xe^x$$

$$\text{Solution: } y = c_1 e^x + c_2 x e^x + \frac{1}{6} x^3 e^x$$

Method of Undetermined Coefficients: Case 2

$$y'' + by' + cy = f(x) \quad f(x) = P_n(x)e^{\alpha x} \cos(\beta x) \text{ or } f(x) = P_n(x)e^{\alpha x} \sin(\beta x)$$

$$\text{Look for } y_p \text{ s.t.} \quad y'' + by' + cy = P_n(x)e^{(\alpha+i\beta)x} \quad [*]$$

Using method in Case 1 to obtain:

$$z(x) = u(x) + iv(x)$$

$$\text{Substitute } z(x) \text{ into } [*] \quad u(x) \text{ is solution of } y'' + by' + cy = P_n(x)e^{\alpha x} \cos(\beta x)$$

$$v(x) \text{ is solution of } y'' + by' + cy = P_n(x)e^{\alpha x} \sin(\beta x)$$

Example

$$y'' - 2y' + 2y = e^x \cos x$$

$$y = c_1 e^x \cos(x) + c_2 e^x \sin(x) + \frac{x e^x \sin x}{2}$$

Announcements

- No homework this week
- Homework for week 2, 5, 9, 10, 11 with each 2% for the final grade
- Read “Guidelines for Mid-term Project” in eDimension
- Next week CNY: No class